

HW #4 Solutions

CSEE W3827 - Fundamentals of Computer Systems Spring 2026

Prof. Rubenstein

Topics: flip-flops, state design, sequential circuit design

Note that this homework has 3 HARDER problems and is 10 pages long.

Warmup Problems

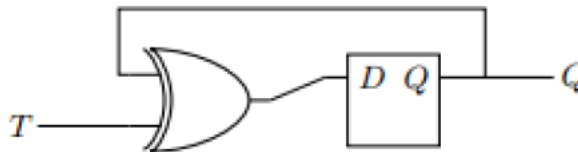
1. Build

- (a) a T flip-flop out of a D flip-flop and combinational circuitry.

Answer: Let's build this circuit such that the output of the D flip-flop is the desired output of the T flip-flop we are attempting to build. First, let's design the state table, whose inputs are the input to the T flip-flop, the current value stored in the D flip-flop, and the desired output (which emulates the T flip-flop).

In	$D(t)$	$D(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

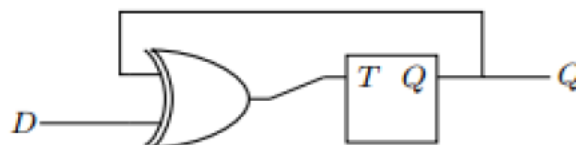
Since the value fed into the D flip-flop at time t will equal its output at time $t+1$, our desired input to the D flip-flop is the right-most column of our state table. Note this value, in terms of inputs and existing states, is simply $In \oplus D(t)$.



- (b) a D flip-flop out of a T flip-flop and combinational circuitry.

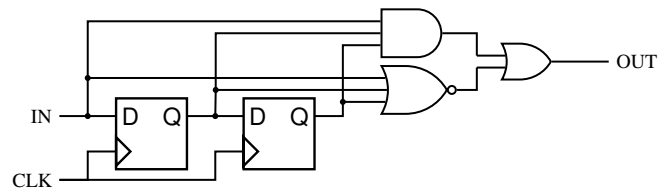
Answer: Using a similar approach as above, we consider the input and current value in the T flip-flop, and determine the next value we need the T flip-flop to hold in order to emulate the behavior of a D flip-flop. Note $T(t+1)$ should just match the input to properly emulate a D flip-flop. Our state table must contain one additional column that shows the value that must be fed into the T flip-flop at time t such that it goes from its value (at time t) to the value it is supposed to hold at time $t+1$.

In	$T(t)$	$T(t+1)$	T_{In}
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0



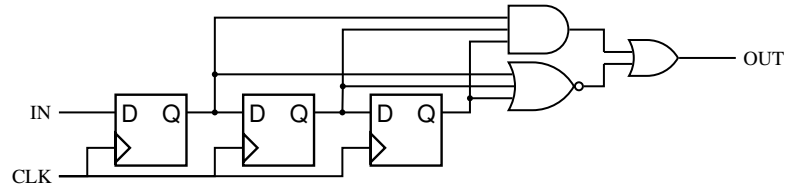
2. Build a sequential circuit that returns 1 whenever the last 3 inputs (INCLUDING the current input) were identical.

Answer:



3. Build a sequential circuit that returns 1 whenever the last 3 inputs (PRIOR to the current input) were identical.

Answer:



Harder Problems

1. A special flip-flop is formed from three latches, arranged in sequence. The first two latches behave as in a normal flip-flop: the leader latch's enable is attached to the clock, and the follower's enable is attached to the complement of the clock. The third latch, which follows the follower, is also attached to the clock. What's different about the outputs of this flip-flop?

Answer: A normal, 2-latch flip flop reads its input during the up-pulse (or down edge) and changes its output on the down pulse (or up-edge). This one delays the output for another half cycle, so that the output changes at the same time the new input is "locked" into the leader latch.

2. Design a circuit using JK flip-flops that takes in a binary stream $B_0B_1B_2B_3 \dots$ and outputs a 1 at time t if the stream received thusfar (i.e., $B_0B_1 \dots B_{t-1}B_t$), when read as an unsigned binary number from low bit to high, is divisible by 3.

The following table depicts a sample input stream and what should be output.

t	0	1	2	3	4	5	6	7	8
$In(t)$	0	0	1	1	0	0	1	0	0
Val of input	0	0	4	12	12	12	76	76	76
val mod 3	0	0	1	0	0	0	1	1	1
output	1	1	0	1	1	1	0	0	0

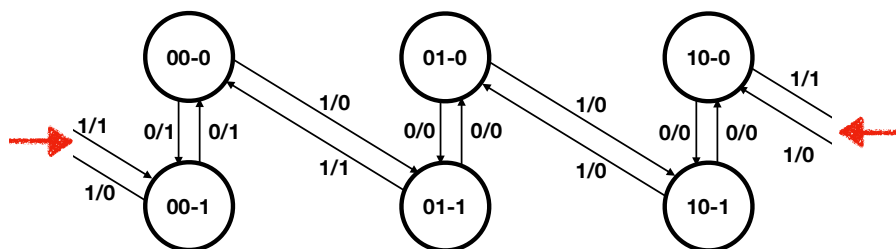
Hint: The value of the new bit starts as equal to 1 which is $1 \bmod 3$, then is 2 which is $2 \bmod 3$, then is 4 which is back to $1 \bmod 3$, etc.

- (a) Draw the state machine, numbering your states in an "obvious" manner. You may assume that the machine starts in the right state at time $t = 0$.

Answer:

This problem can actually be solved using only 3 states, but the explanation is a bit complex, so suffice it to solve this problem using 6 states.

Each state tracks two pieces of information: the first is the current value modulo 3 (we obviously have to track this value since it determines the output). The second piece of information is the value to be added when the current input is a 1. An important observation is that the i th input bit being 1 which adds 2^i to the overall value changes the value mod 3 by 1 when i is even, and by 2 when i is odd. Thus, we have to track whether the current timeslot is an even timeslot or an odd timeslot. Hence, a state will track the current value mod 2 (0, 1, or 2) and whether the current clock cycle is an even (0) or odd (1) clock cycle.



Note in the above state machine, each state ABC is represented as $AB - C$ to distinguish the bits AB that represent the current value mod 3 (not including the current input), and C represents the value 2^C that the current input has mod 3. Note that in this state machine, we output a 1 when the value that *includes the current input* equals $0 \bmod 3$, i.e., we output 1 when transitioning to a state where $AB = 00$.

Note also that the red arrows on the margins indicate where the transitions wrap across the page.

- (b) Give the algebraic formula (simplified as much as possible) for the J and K inputs of your flip-flops, and also for the output.

Answer:

time t					$t + 1$	time t			$t + 1$	time t			$t + 1$	time t		
A	B	C	In	Out	A	J_A	K_A	B	J_B	K_B	C	J_C	K_C	A	B	C
0	0	0	0	1	0	0	X	0	0	X	1	1	X	0	0	0
0	0	0	1	0	0	0	X	1	1	X	1	1	X	0	0	1
0	0	1	0	1	0	0	X	0	0	X	0	X	1	0	0	0
0	0	1	1	0	1	1	X	0	0	X	0	X	1	0	0	1
0	1	0	0	0	0	0	X	1	X	0	1	1	X	0	1	0
0	1	0	1	0	1	1	X	0	X	1	1	1	X	0	1	1
0	1	1	0	0	0	0	X	1	X	0	0	X	1	0	1	0
0	1	1	1	1	0	0	X	0	X	1	0	X	1	0	1	1
1	0	0	0	0	1	X	0	0	0	X	1	1	X	1	0	0
1	0	0	1	1	0	X	1	0	0	X	1	1	X	1	0	1
1	0	1	0	0	1	X	0	0	0	X	0	X	1	1	0	0
1	0	1	1	0	0	X	1	1	1	X	0	X	1	1	0	1
1	1	0	0	X	X	X	X	X	X	X	X	X	X	1	1	0
1	1	0	1	X	X	X	X	X	X	X	X	X	X	1	1	1
1	1	1	0	X	X	X	X	X	X	X	X	X	X	1	0	0
1	1	1	1	X	X	X	X	X	X	X	X	X	X	1	1	1

Answer continued on next page.

Answer:

Continued from previous page

AB \ C In				
	00	01	11	10
00	0	0	1	0
01	0	1	0	0
11	X	X	X	X
10	X	X	X	X

AB \ C In				
	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	X	X	X	X
10	0	1	1	0

$$J_A = B\bar{C}In + \bar{B}CIn$$

$$K_A = In$$

AB \ C In				
	00	01	11	10
00	0	1	0	0
01	X	X	X	X
11	X	X	X	X
10	0	0	1	0

AB \ C In				
	00	01	11	10
00	X	X	X	X
01	0	1	1	0
11	X	X	X	X
10	X	X	X	X

$$J_B = \bar{A}\bar{C}In + ACIn$$

$$K_B = In$$

Obviously: $J_C = K_C = 1$.

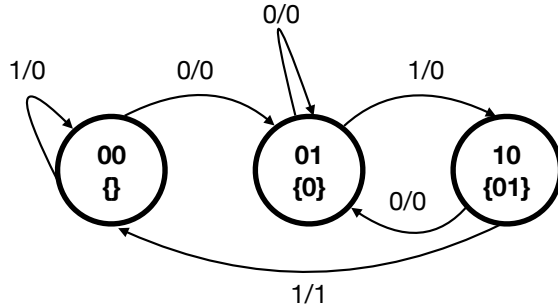
AB \ C In				
	00	01	11	10
00	1	0	0	1
01	0	0	1	0
11	X	X	X	X
10	0	1	0	0

$$Out = A\bar{C}In + BCIn + \bar{A}\bar{B}\bar{I}n$$

3. Consider a sequential circuit that reads in an input $X(t)$ during clock cycle t . The circuit should look for the sequence 011. Design the sequential circuit (i.e., give simplified algebraic expressions) using D flip-flops for each of the following versions:
- (a) If the second 1 arrives during clock cycle t , then the circuit should output a 1 during clock cycle t , and otherwise outputs a 0.

Answer:

To design this state machine, there needs to be a state that indicates that a first 0 has been received, then a state that indicates that thusfar 01 has been received. When in the 01 state, if a 1 is received, the machine can output 1. Last, there must also be state that indicates that the sequence has not yet been initiated (i.e., a 1 has been received without a 0 immediately preceding it or without a 01 immediately preceding it. Since 3 states are needed, each state needs to be described by (at minimum) a 2-bit sequence, $A(t)B(t)$. Arbitrarily, we will assign 00 to the not-yet-initiated state, 01 to the received-0 state, and 10 to the received 01 thusfar.



$A(t)$	$B(t)$	$In(t)$	$Out(t)$	$A(t+1)$	$B(t+1)$
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	0	0	1
0	1	1	0	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	X	X	X
1	1	1	X	X	X

A	B In			
	00	01	11	10
0	0	0	0	0
1	0	1	X	X

$$Out(t) = A(t)In(t)$$

A	B In			
	00	01	11	10
0	0	0	1	0
1	0	0	X	X

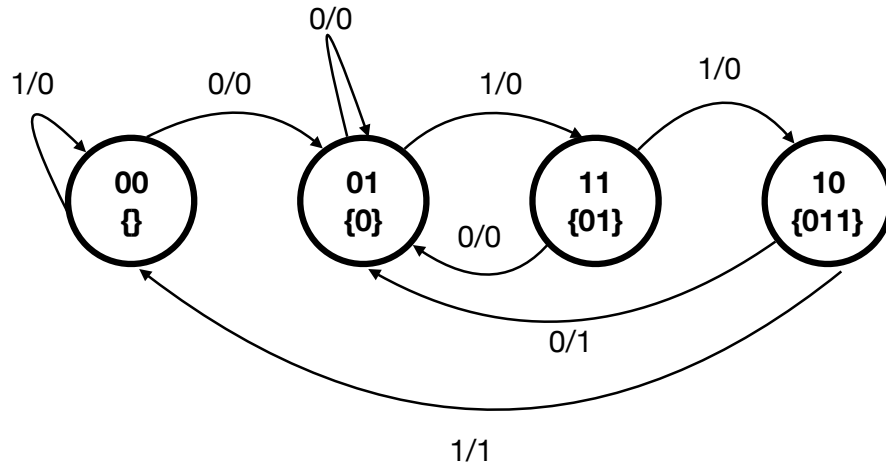
$$A(t+1) = B(t)In(t)$$

A	B In			
	00	01	11	10
0	1	0	0	1
1	1	0	X	X

$$B(t+1) = \overline{In(t)}$$

- (b) If the second 1 arrives during clock cycle t , then the circuit should output a 1 during the clock cycle $t + 1$, and otherwise outputs a 0.

Answer: Here, we need 4 states. One that represents the sequence not yet received at all (we will use state 00). Another for the 0 having been received (we will use state 01), another for 01 received (we will use state 11) and another for 011 received (we will use state 10).



$A(t)$	$B(t)$	$In(t)$	$Out(t)$	$A(t+1)$	$B(t+1)$
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	0	0	1
0	1	1	0	1	1
1	0	0	1	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	0	1	0

A \ B In				
	00	01	11	10
0	0	0	0	0
1	1	1	0	0

$$Out(t) = A(t)\overline{B(t)}$$

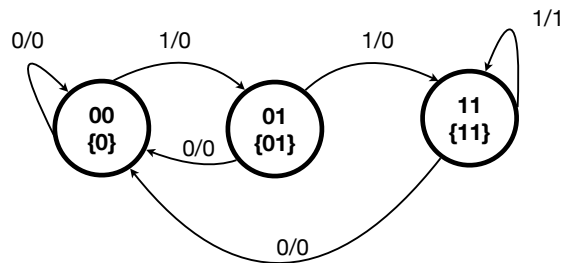
A \ B In				
	00	01	11	10
0	0	0	1	0
1	0	0	1	0

$$A(t+1) = B(t)In(t)$$

A \ B In				
	00	01	11	10
0	1	0	1	1
1	1	0	0	1

$$B(t+1) = \overline{I(t)} + \overline{A(t)}B(t)$$

- (c) Suppose the sequence is 111 instead of 011. If at time t , the current and two previous inputs were all 1, then a 1 should be output during clock cycle t , and otherwise a 0 should be output. Note that multiple consecutive outputs of 1 are possible.



Answer:

A	B	In	$A' = D_A$	$B' = D_B$	Out
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	1	0
1	0	0	X	X	X
1	0	1	X	X	X
1	1	0	0	0	0
1	1	1	1	1	1

A	B In			
	00	01	11	10
0	0	0	1	0
1	X	X	1	0

$$D_A = BIn$$

A	B In			
	00	01	11	10
0	0	1	1	0
1	X	X	1	0

$$D_B = In$$

A	B In			
	00	01	11	10
0	0	0	0	0
1	X	X	1	0

$$Out = AIn$$